# MIZORAM PUBLIC SERVICE COMMISSION

## TECHNICAL COMPETITIVE EXAMINATION FOR JUNIOR GRADE OF MIZORAM HEALTH SERVICES (COMBINED) (DENTAL SURGEON SUB-CADRE) UNDER HEALTH & FAMILY WELFARE DEPARTMENT, GOVERNMENT OF MIZORAM. (SEPTEMBER-2025)

## TECHNICAL SUBJECT PAPER-III (DENTAL SURGEON)

Time Allowed : 2 hours FM : 200

*All questions carry equal marks of 2 each.*
*Attempt all questions.*

**1.** Which of the following is the most important drug in the treatment of anaphylactic shock?
(a) Corticosteroids (b) Adrenaline
(c) Antihistamines (d) Sodium bicarbonate

**2.** Which of the following is effective against anaerobic bacteria?
(a) Metronidazole (b) Ofloxacin
(c) Cefixime (d) All of the above

**3.** Which of the following analgesics is contraindicated in patients with liver disease?
(a) Acetaminophen (b) Ibuprofen
(c) Paracetamol (d) Both (a) and (b)

**4.** What is the primary indication for administering epinephrine in a dental office?
(a) Cardiac arrest (b) Hypotension
(c) Anaphylaxis (d) Hypertension

**5.** One of your patient, Rinzuala, stated that he is narcotic addict. After the extraction which of the following analgesics are contraindicated?
(a) Aspirin (b) Acetaminophen
(c) Pentazocin (d) None of the above

**6.** The material recommended for root surface caries is-
(a) GIC (b) Direct filling gold
(c) Composites (d) Amalgam

**7.** While giving PSA nerve block you forgot to aspirate before injecting and soon after administering LA, a swelling developed in the posterior maxilla which is visible extraorally with no other signs and symptoms. The swelling is probably is due to:
(a) Allergy (b) Infection
(c) Overdose (d) Hematoma

**8.** The ideal rate for administering LA is:
(a) 1.8 ml/min (b) 1.1 ml/min
(c) 2 ml/min (d) 1 ml/min

**9.** Smaller diameter implant needs to be placed in:
(a) Division A bone (b) Division B bone
(c) Division C bone (d) Division D bone

**10.** _____class of drugs are currently the best oral sedative drugs for dentistry.
(a) Narcotics (b) Barbiturates
(c) Benzodiazepines (d) Ethanols

**11.** Intra-articular injections of steroids into painful arthritic TMJ have palliative effects from:
(a) Providing an analgesic effect (b) Providing an anti inflammatory action
(c) Providing better lubricating effect (d) Increasing blood supply

**12.** Amide type local anaesthetics are metabolized in:
(a) Serum (b) Spleen
(c) Liver (d) Kidney

**13.** Instead of responding to treatment for syncope, the patient's pulse and respiration become weak and irregular with attendant cyanosis. The first resuscitative measure should be to-
(a) Support circulation by injecting 1:1000 adrenaline
(b) Begin closed chest cardiac massage
(c) Support respiration with oxygen through open airway
(d) Place a paper bag over patient's face and raised blood CO2

**14.** Which number scalpel blade is universally useful for oral surgical procedures?
(a) 15 (b) 17
(c) 12 (d) 10

**15.** Pre-operative vitamin K is indicated in patients with:
(a) Diabetes mellitus (b) Liver disease
(c) Chronic pneumonitis (d) Scurvy

**16.** Osseo integration implies:
(a) The process of inserting implants in bone
(b) Integration of the bone and implant with fibrous tissue between them
(c) Direct contact of living bone and the surface of the implant at the light microscopic level of magnification
(d) All of the above

**17.** The most common type of implant in use today are:
(a) Subperiosteal implant (b) Endosteal implant
(c) Transosteal implant (d) All of the above

**18.** The minimum space between implant should be:
(a) 1 mm (b) 2 mm
(c) 3 mm (d) 5 mm

**19.** Time taken for integration of implants in the maxilla is :
(a) 4 months (b) 6 months
(c) 2 months (d) 3 months

**20.** The recommended time interval between surgery and placing load in the posterior Mandible is:
(a) 2 months (b) 6 months
(c) 4 months (c) 8 months

**21.** What is the primary factor affecting implant stability?
(a) Bone density (b) Surgical technique
(c) Implant design (d) All of the above

**22.** Which of the following Interpretation of ISQ values is for high implant stability?
(a) 20-30 (b) 60-65
(c) 40-50 (d) > 65

**23.** Which type of laser is commonly used for soft tissue procedures in dentistry?
(a) Nd:YAG laser (b) Er:YAG laser
(c) CO2 laser (d) Diode laser

**24.** Which of the following procedures can be performed using lasers in dentistrt?
(a) Teeth whitening (b) Cavity preparation
(c) Gum reshaping (d) All of the above

**25.** Which laser wavelength is commonly used for hard tissue procedures in dentistry?
(a) 2940 nm ( Er:YAG) (b) 810 nm ( Diode)
(c) 1064 nm( Nd:YAG) (d) 10,600 nm ( CO2)

**26.** The absolute contraindication for use of adrenalin in LA is:
(a) Myocardial infarction 6 months ago (b) Hyperthyroidism
(c) Angina pectoris (d) Pregnancy

**27.** Which of the following is a long acting LA agent?
(a) Mepivacaine (b) Prilocaine
(c) Bupivacaine (d) Propoxycaine

**28.** The most important design characteristic of a bur blade is:
(a) Rake angle (b) Clearance angle
(c) Edge angle (d) Taper angle

**29.** The most widely accepted intracanal irrigant displaying optimal cleansing and bacteriocidal properties is:
(a) Formocresol (b) Sterile saline
(c) Hydrogen peroxide (d) Sodium hypochlorite

**30.** The following is not a component of fixed appliance:
(a) Bands (b) Adam's clasp
(c) Brackets (d) Open hellicle

**31.** The impression material recommended for taking impression for maxillofacial prosthesis is:
(a) Alginate (b) Silicone
(c) ZnOE (d) POP

**32.** Replanted avulsed tooth can be aesthetically stabilized by:
(a) Occlusal splints (b) Arch bar
(c) Composite resins (d) All of the above

**33.** Instrument that is more likely to break in root canal when twisted more than 90 degrees.
(a) File (b) Spreader
(c) Broach (d) Reamer

**34.** Instrument of choice for bio mechanical preparation in fine root canals of double curved canals.
(a) H file (b) Unifile
(c) K-flex file (d) Reamer

**35.** Which of the following materials is commonly used for making temporary crowns?
(a) Acrylic resin (b) Ceramic
(c) Composite resin (d) Metal

**36.** What is the primary functions of apex locator in endodontics?
(a) To measure the length of the root canal
(b) To determine the working length of the root canal
(c) To detect the presence of bacteria in the root canal
(d) To assess the quality of the root canal filling

**37.** What is the purpose of a periodontal probe?
(a) To measure the depth of periodontal pockets
(b) To detect calculus on teeth
(c) To remove plaque from teeth
(d) To apply periodontal dressing

**38.** Cavity varnish is used for:
(a) To reduce thermal conductivity
(b) To reduce the permeability of acid into dentinal tubules
(c) Action as an obtundant
(d) All of the above

**39.** Which of the following is common to both zinc eugenol cement and polycarboxylate cement?
(a) Polyacrylic acid is liquid
(b) Chemical bond to tooth structure
(c) Chelation
(d) Substitution of eugenol by EBA to increase strength of cement

**40.** Which of the following cements is most biocompatible with pulp?
(a) GIC
(b) Copper cement
(c) Zinc phosphate cement
(d) Zinc oxide eugenol cement

**41.** Thermal conductivity of which material is close to dentin :
(a) Amalgam
(b) Silicate
(c) Composite resin
(d) Cavity varnish

**42.** Bond found in GIC is:
(a) Covalent bond
(b) Ionic bond
(c) Hydrogen bond
(d) Metallic bond

**43.** Creep rate decreases with:
(a) Increase force of condensation
(b) Decrease force of condensation
(c) Under or overtrituration
(d) Delay in time between trituration and condensation

**44.** The carat for gold foil used for restoration is:
(a) 18
(b) 20
(c) 22
(d) 24

**45.** The chemical used to etch enamel is :
(a) Zinc oxide
(b) Methyl methacrylate
(c) Phosphoric acid
(d) Eugenol

**46.** Shade selection is done with in how much time to avoid fatigue of eyes?
(a) 5 seconds
(b) 10 seconds
(c) 15 seconds
(d) 20 seconds

**47.** Sodium nitrite is used as:
(a) Antiplaque agent (b) Antirust agent
(c) Desensitising agent (d) Bleaching agent

**48.** The matrix retainer where in anatomy adaptation is possible without wedges is:
(a) Tofflemire (b) Steele's Siqueland
(c) Ivory no.1 (d) Ivory no. 8

**49.** Maximum dose of xylocaine with out adrenaline that can be given in 60kg adult is :
(a) 500 mg (b) 300 mg
(c) 400 mg (d) 600 mg

**50.** Most commonly used local anesthesia:
(a) 1:20,000 xylocaine Hcl (b) 1:50,000 xylocaine Hcl
(c) 1:10,000 xylocaine Hcl (d) 1:80,000 xylocaine Hcl

**51.** Which of the following is primarily a bactericidal drug?
(a) Chloramphenicol (b) Gentamicin
(c) Sulphadiazine (d) Tetracycline

**52.** Tetracycline is contraindicated in :
(a) Children (b) Pregnancy
(c) Renal failure (d) All of the above

**53.** Omeprazole is a:
(a) Antihistamine (b) Proton pump inhibitor
(c) Ulcer protective (d) Ulcer healing drug

**54.** In a patient who has come for dental extraction, having asthama is on theophylline you better avoid:
(a) Lignocaine (b) Azithromycin
(c) Ciprofloxacin (d) Metronidazole

**55.** Drugs highly bound to albumin.
(a) NSAID'S (b) Lidocaine
(c) Beta blockers (d) Verapamil

**56.** Which of the following can be given safely to a patient of congestive heart failure?
(a) Aspirin (b) Paracetamol
(c) Diclofenac sodium (d) Ibuprofen

**57.** Pregnant patient can be safely given :
(a) Tetracycline (b) Paracetamol
(c) Metronidazole (d) Barbiturates

**58.** NSAID'S have adverse effect on:
(a) Bone (b) Kidney
(c) Liver (d) Stomach

**59.** Route of choice for analgesics in cancer patients:
(a) Oral (b) Iv
(c) Topical (d) Sublingual

**60.** The drug of choice in the management of life threatening allergic reaction is:
(a) Corticosteroids (b) Antihistamines
(c) Adrenalin (d) Diazepam

**61.** Bleeding from the oral mucosa can be reduced by topical use of:
(a) Epinephrine (b) Lidocaine
(c) Hydrocortisone (d) Dicoumarol

**62.** A common side effect associated with all NSAID drugs is:
(a) Drowsiness (b) Gastric irritation
(c) Xerostomia (d) Constipation

**63.** After the first implant is placed anterior to the mental foramen, the second dental implant should be placed:
(a) In the symphysis area
(b) Anterior to the mental foramen on the opposite side
(c) 3 mm away from the first implant
(d) 5 mm away from the first implant

**64.** Implants are most often made of:
(a) Titanium (b) Hydroxyapatite
(c) Stainless steel (d) Gold alloy

**65.** Most plausible factor for dental implant fracture:
(a) Peri – implantitis (b) Bending overloads
(c) Non – passive fit prosthesis (d) Manufacturing imperfections

**66.** A non invasive way of evaluating implant stability by impact resistance:
(a) Periotest (b) Transgingival probing
(c) Bone sounding (d) Probing

**67.** Hydroxyapatite coating of dental implants was introduced by :
(a) DeGroot (b) Hahn and Polich
(c) Kay (d) Clemow

**68.** Helium –neon (HeNe) lasers are example for :
(a) Excimer lasers (b) Gas lasers
(c) Diode lasers (d) Solid state lasers

**69.** The unalloyed form of titanium used to form dental implants is in the :
(a) Beta form (b) Alpha form
(c) Gamma form (d) Delta form

**70.** Screw – retained prosthesis possess the following advantages:
(a) Ease of fabrication (b) Ease of retrieval
(c) Esthetics (d) Ease of connection to natural teeth

**71.** Immediate implant placement is :
(a) The placement of an implant at the time of tooth extraction
(b) The placement of multiple implants at once
(c) The placement of a restoration at the time of implant placement
(d) The placement of a restoration at the time of surgical uncovering.

**72.** Dense cortical bone used for implant placement is found in:
(a) Anterior mandible (b) Posterior mandible
(c) Anterior maxilla (d) Posterior maxilla

**73.** The ideal bone for implant support is:
(a) Spongy bone (b) Cortical bone
(c) Lamellar bone (d) Woven bone

**74.** Ambu bag is :

(a) Bag valve mask device
(b) Pocket mask
(c) Positive pressure oxygen cylinder
(d) None of the above

**75.** The ideal technique for emergency drug administration is:

(a) Intravenous
(b) Intramuscular
(c) Subcutaneous
(d) Endotracheal

***Direction (Question Nos. 76 - 79) : Look at the word in capital letters. Then, from the four words that follow, find the one that means the same thing as the word in capitals:***

**76.** WHIMSICAL

(a) unlike
(b) musical
(c) dancing
(d) unpredictable

**77.** APATHETIC

(a) ignorant
(b) indifferent
(c) pitiful
(d) concerned

**78.** CLANDESTINE

(a) furtive
(b) safe
(c) tribal
(d) open

**79.** INCULCATE

(a) grow
(b) inquire
(c) instill
(d) compute

**80.** First read the saying or proverb in capital letters. Then, from the five sayings under it, find the one that means the same thing, and mark your answer:

LET ANOTHER'S SHIPWRECK BE YOUR GUIDE AT SEA

(a) one man's meat is another man's poison
(b) ships fear fire more than water
(c) wise men learn from other men's mistakes
(d) the worst things happen at sea

**81.** Please select a pair of words that have similar relationships

Flower – bouquet

(a) soldier – company
(b) bread – cookie
(c) magazine – cover
(d) tap – water

**82.** Find the odd one out from the following anagrams:

(a) SCHAMOT
(b) LABLOTOF
(c) CEKTIRC
(d) SNINET

***Direction (Question No. 83) : Pick out the correct synonym of the word underlined:***

**83.** The police officer found that the documents recovered were <u>fabricated.</u>

(a) constricted
(b) genuine
(c) forged
(d) prepared

***Directions (Question No. 84) : Find the Missing number in the series given below:***

**84.** 121, 112, ?, 97, 91, 86

(a) 102 (b) 108
(c) 99 (d) 104

**85.** Among four playing cards, a Heart is to the immediate left of a spade, a 9 is to the left of a J, a Diamond to the right of a Heart, a 7 is the left most card and a Club is the right most card. An 8 is to the left of a 9. Then the 9 is a _____.

(a) Spade (b) Heart
(c) Diamond (d) Club

***Direction (Question Nos. 86 & 87) : Given below are four sets of letters. Three of the four sets follow a certain rule. One set does not. Mark the letter of the one set that does not follow the rule:***

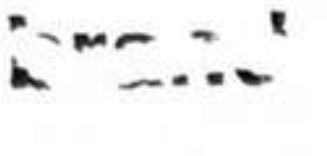
**86.**

(a) CDED (b) LMOM
(c) PQRQ (d) STUT

**87.**

(a) DECG (b) JKIL
(c) MNLO (d) QRPS

***Direction (Question Nos. 88 & 89) : In this test, you will see a word with parts of the letters missing. This "incomplete" word is followed by four "jumbled" words. You are first to use your imagination to figure out what the incomplete word is. Then find, among the four scrambled choices, the one that has the right letters to spell the incomplete word:***

**88.** 

(a) rodba (b) dnbra
(c) aibnr (d) radeb

**89.** 

(a) lowilw (b) rdweno
(c) dwowni (d) nawred

**90.** The average height of students in a class of 10 is 105cm. If 20 more students with an average height of 120cm join the class, what will the average height be?

(a) 105 cm (b) 110 cm
(c) 112 cm (d) 115 cm

**91.** A student has to secure 35% marks to pass. He got 80 marks and failed by 60 marks. The maximum marks are:

(a) 100 (b) 200
(c) 300 (d) 400

**92.** 125% of 32 = _____

(a) 6 (b) 12
(c) 36 (d) 40

**93.** Which has the longest perimeter?

(a) a square of side 10 cms
(b) a rectangle of sides 12 cms and 9 cms
(c) a circle of radius 7 cms
(d) a rhombus of side 9 cms

**94.** $\frac{5}{7}$ of 245 is:

(a) 165 (b) 175
(c) 207 (d) 343

**95.** Which one of the numbers below could replace X in both places in 4 + X = 6 - X?

(a) 4 (b) 3
(c) 2 (d) 1

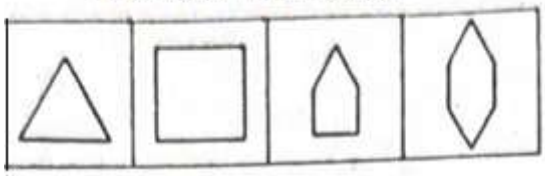
**96.** The third smallest 3-digit odd number using the digits 1 to 9 is ___.

(a) 105 (b) 123
(c) 113 (d) 115

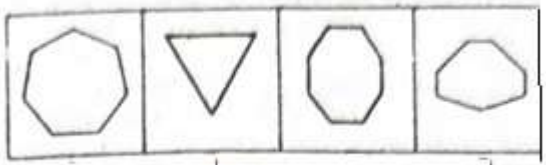
***Direction (Question Nos. 97 - 99) : In this test you will see rows of designs or figures. Each row has four designs called Problem figures. These are followed by four designs called Answer figures. The four Problem figures make a series. That means, they follow each other in a special order. The fifth figure is missing. Find the right picture among the four Answer figures and mark the correct one:***

**97.**

**PROBLEM FIGURES**



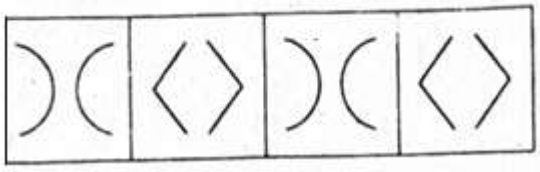
**ANSWER FIGURES**



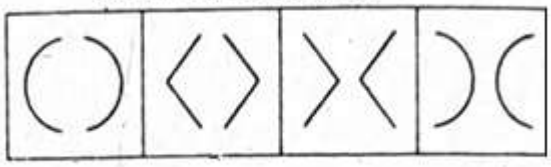
(a) (b) (c) (d)

**98.**

PROBLEM FIGURES



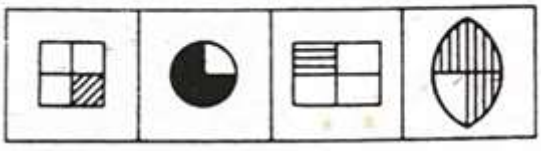
ANSWER FIGURES



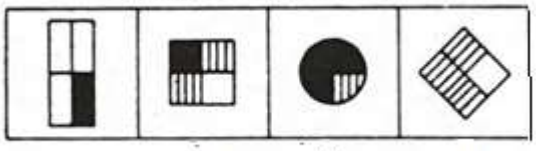
(a) (b) (c) (d)

**99.**

PROBLEM FIGURES

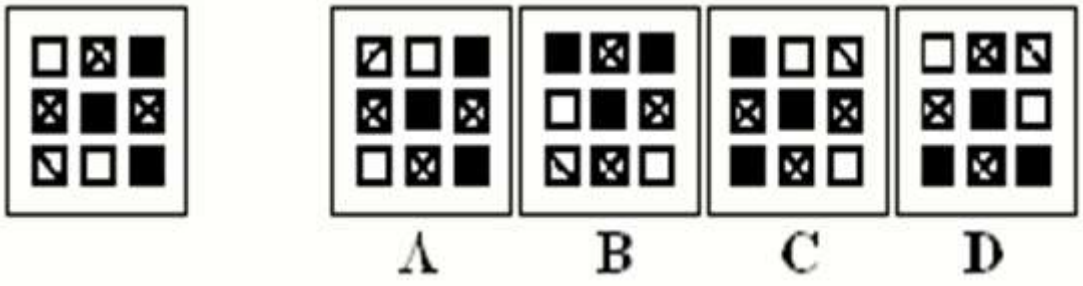


ANSWER FIGURES



(a) (b) (c) (d)

**100.** In the figures given below, one of the shapes (A –D) is identical to the first figure, but has been rotated. Which figure is identical to the first?




(a) A
(b) B
(c) C
(d) D

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